Supplemental Material: Yukawa screening derivation of the bond-valence rule

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DATA PROVENANCE

The figures and tables in this Supplemental Material cover the same 103 cation–oxygen species analyzed in the main text, spanning *s*-, *p*-, *d*-, and *f*-block cations.

Source structures are DFT-relaxed entries from the Materials Project. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph, and (R_0, B) were determined by linear least squares. For rank-deficient bond graphs (fewer independent bonds than parameters), an anion-centered reparameterization was used; any remaining ill-conditioned cases were resolved with the constrained nonlinear procedure of Ref. 1, with R_0 bounds $(-10, 10)$ Å. Coordination numbers were determined from the bond graph.

The DFT-centroid comparison in Fig. 2 uses ten *s*-block oxides in panel a (Li_2O, Na_2O, K_2O, Rb_2O, Cs_2O, BeO, MgO, CaO, SrO, and BaO) and 25 non-*s*-block oxides in panel b: Al_2O_3, SiO_2, GeO_2, SnO_2, PbO, TiO_2, V_2O_5, CoO, NiO, CuO, ZnO, ZrO_2, MoO_3, PdO, Ag_2O, HfO_2, Ta_2O_5, Au_2O_3, La_2O_3, Gd_2O_3, Yb_2O_3, B_2O_3, P_2O_5, As_2O_3, and Sb_2O_3. The panel-*b* regression is defined a priori as a test of the Yukawa screening picture only for ground-state binary oxides whose cation environments approximate an isotropic first-shell ionic screening cloud. It therefore uses the 21 filled-symbol oxides and excludes B_2O_3, P_2O_5, As_2O_3, and Au_2O_3, whose ground-state structures are dominated by directional, non-isotropic bonding motifs; those four compounds are shown as open symbols. The 21 oxides shown as filled symbols in Fig. 2(b) are the 25 listed above minus the four directional cases just named; no further data-driven exclusion was applied beyond this fixed list. The excluded oxides are the cases in this benchmark whose ground-state cation environments are not compact first-shell ionic polyhedra but directional or terminal-oxyanion motifs, so a single shell centroid m_i is not an adequate descriptor of the screening cloud.

COORDINATION-NUMBER-RESOLVED B–R_0 FIT LINES

Figures 1–15 show the n-resolved B vs. R_0 fit lines for every species in the 103-cation set, grouped by block: Group 1 (Fig. 1), Group 2 (Fig. 2), $3d/4d/5d$ transition metals (Figs. 3–10), *p*-block (Figs. 11–13), and *f*-block lanthanides plus actinides (Figs. 14–15). Within each coordination shell, the fitted parameters lie on a linear family $B = \beta^{(n)} R_0 + \beta_0^{(n)}$, and the lines converge toward cation-specific characteristic pairs (R_0^*, B^*) as described in the main text. Together, these atlas pages display every coordination-resolved fit that contributes to the pole form $\beta = 1/\ln(z/n)$ shown in Fig. 1 of the main text.

Weighting scheme for the goodness-of-fit metrics. The weighted R^2 and mean absolute error reported with Fig. 1 of the main text use weights $w_i = \max(N_i, 1)$, where N_i is the number of DFT-relaxed structures contributing to the linear B–R_0 fit at species i and coordination n. This gives more populous (species, n) cells more leverage, in proportion to the number of independent structures backing each fitted slope β_i. Letting $\hat{\beta}_i = 1/\ln(z_i/n)$ be the parameter-free closed-form prediction and $\bar{\beta}_w = \sum_i w_i \beta_i / \sum_i w_i$, the reported metrics are

$$R_w^2 = 1 - \frac{\sum_i w_i \, (\beta_i - \hat{\beta}_i)^2}{\sum_i w_i \, (\beta_i - \bar{\beta}_w)^2}, \qquad \mathrm{MAE}_w = \frac{\sum_i w_i \, |\beta_i - \hat{\beta}_i|}{\sum_i w_i}.$$

Pooling 68 species at $n = 4$ ($\sum w = 13{,}370$) with 82 species at $n = 6$ ($\sum w = 17{,}162$) – the $z \neq n$ subset shown in Fig. 1 – gives $R_w^2 = 0.986$ and $\mathrm{MAE}_w = 0.15$ for the parameter-free closed form $\beta = 1/\ln(z/n)$. Per panel: $R_w^2 = 0.973$, $\mathrm{MAE}_w = 0.23$ at $n = 4$; $R_w^2 = 0.979$, $\mathrm{MAE}_w = 0.089$ at $n = 6$.

Exclusion of $z = n$ species from Fig. 1. Species at $z = n$ (U^{4+}, Mn^{4+}, Ti^{4+}, V^{4+}, Zr^{4+}, Ir^{4+}, P^{4+}, Pb^{4+}, and Te^{4+} for $n = 4$; Mo^{6+} and W^{6+} for $n = 6$) are omitted from Fig. 1 of the main text. The valence-sum rule at $z = n$ enforces $S_{ij} = 1$ for every bond, so $\exp[(R_0 - R_{ij})/B] = 1$ identically; in the symmetric-shell limit this is satisfied for any B

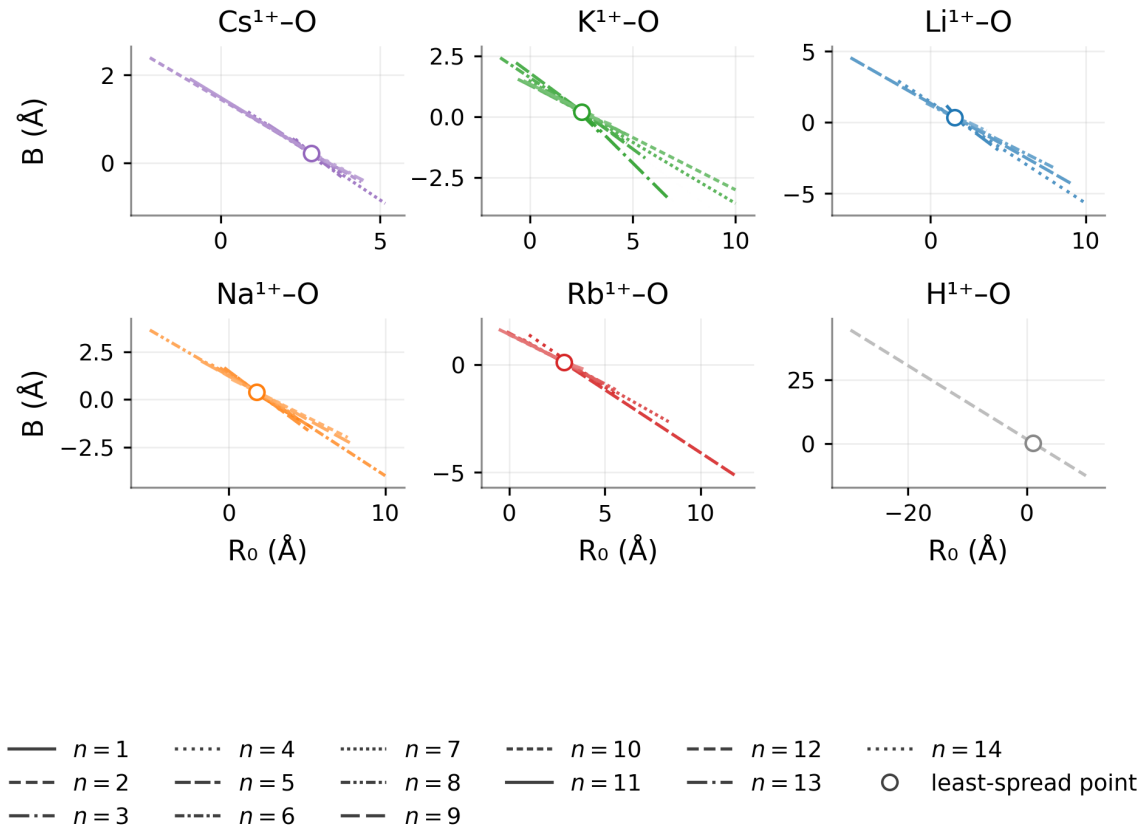

FIG. 1. n-resolved B vs. R_0 fit lines for Group 1 X–O pairs (H, Li, Na, K, Rb, Cs). Each colored line is a different coordination number. The convergence of the lines defines the characteristic pair (R_0^*, B^*) for each cation.

with $R_0 = \bar{R}$, leaving the B–R_0 slope β formally unconstrained. Even with the slight bond-length distortions present in real shells, the resulting fits exhibit very large variance (see the $z = n$ panels in Figs. 3–10 and 11–15, where the coordination-resolved lines have widely scattered slopes and intercepts). The closed-form expression $\beta = 1/\ln(z/n)$ correctly diverges at $z = n$, and the divergence itself is a quantitative prediction of the Yukawa screening picture; we plot only $z \neq n$ species so that finite, well-conditioned fitted slopes are not visually conflated with the formal pole. Fig. 1 therefore shows the 94 species (150 fitted slopes) for which $z \neq n$ at $n = 4$ or $n = 6$. Six species (Cl^{1-}, C^{4+}, Hg^{1+}, H^{1+}, N^{3+}, S^{4+}) lack a fittable line at either $n = 4$ or $n = 6$, P^{4+} lies exactly at the $n = 4$ pole, and Te^{6+} and U^{6+} fall below the $R^2 \geq 0.3$ acceptance threshold at $n = 6$.

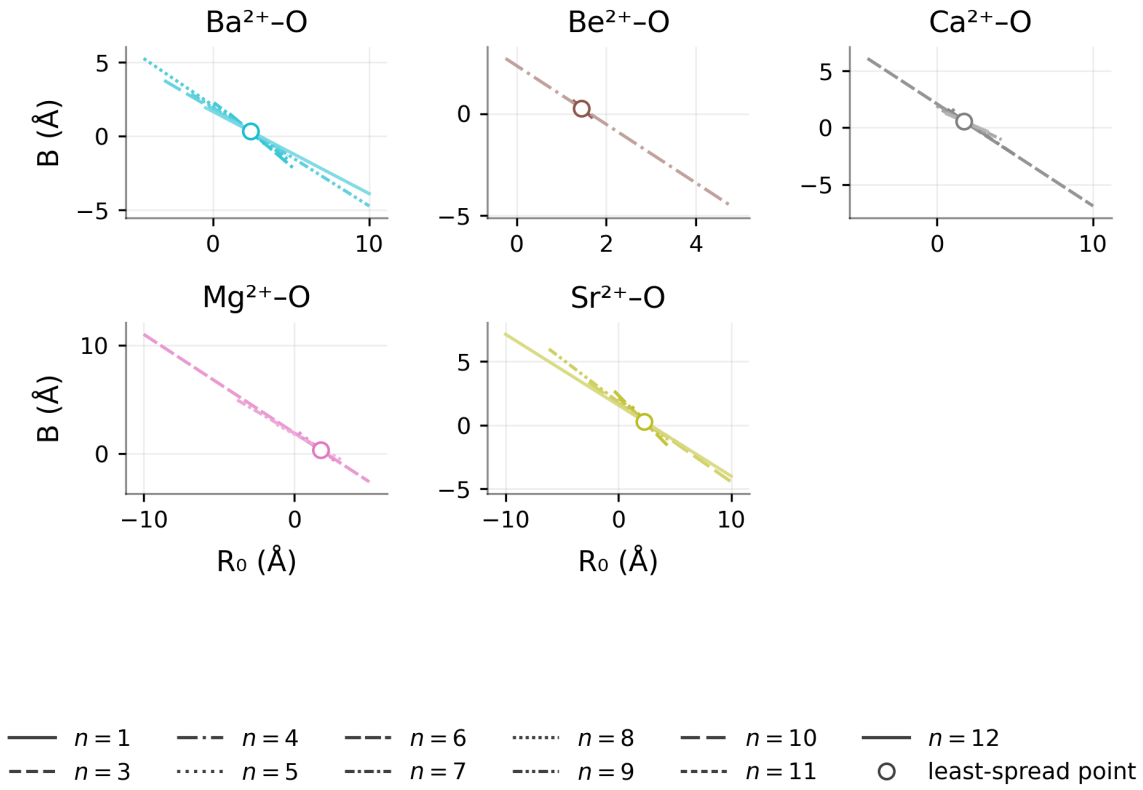

FIG. 2. n-resolved B vs. R_0 fit lines for Group 2 X–O pairs (Be, Mg, Ca, Sr, Ba).

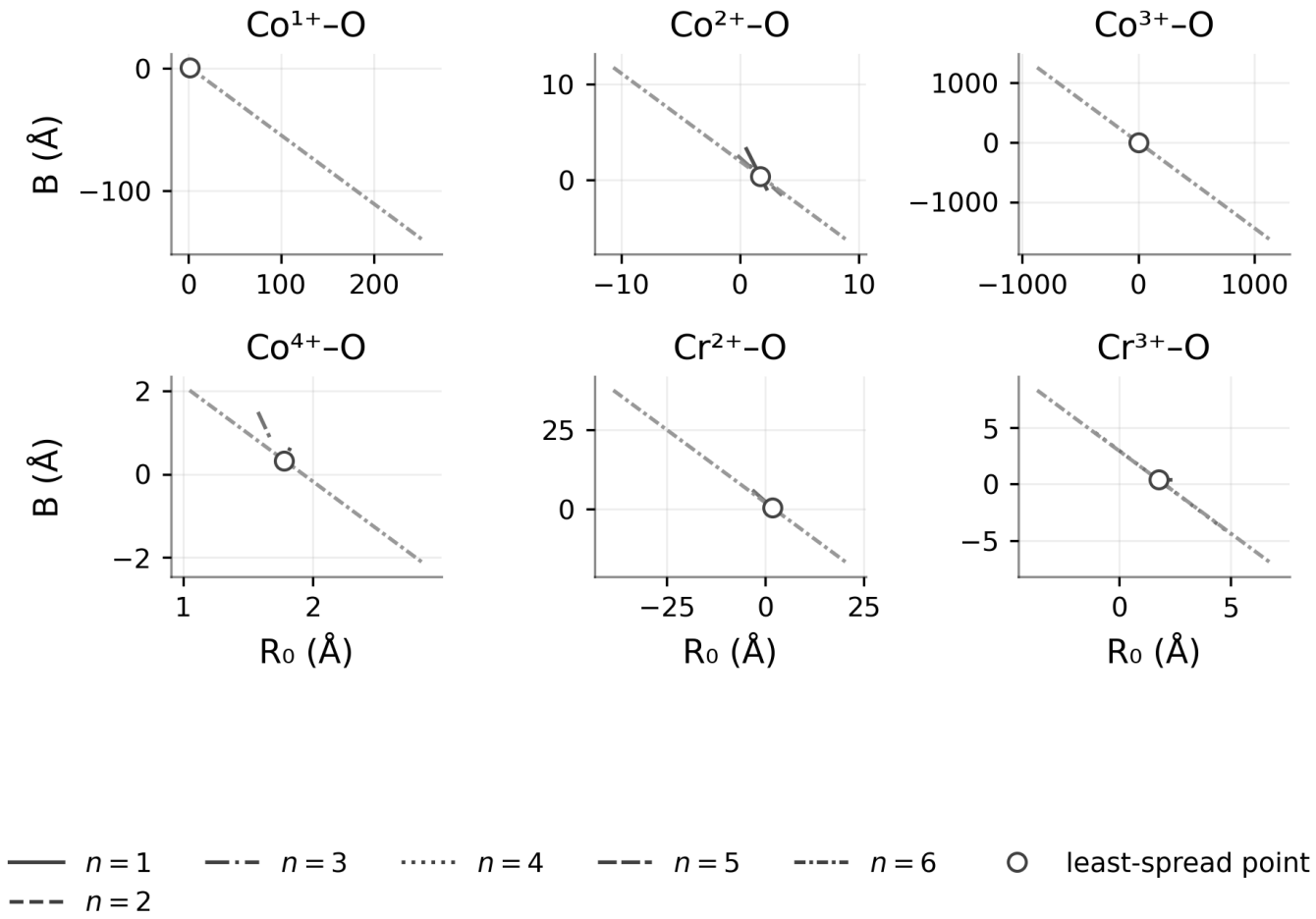

FIG. 3. n-resolved B vs. R_0 for oxidation-state–resolved $3d$ X–O pairs (Sc–Cr).

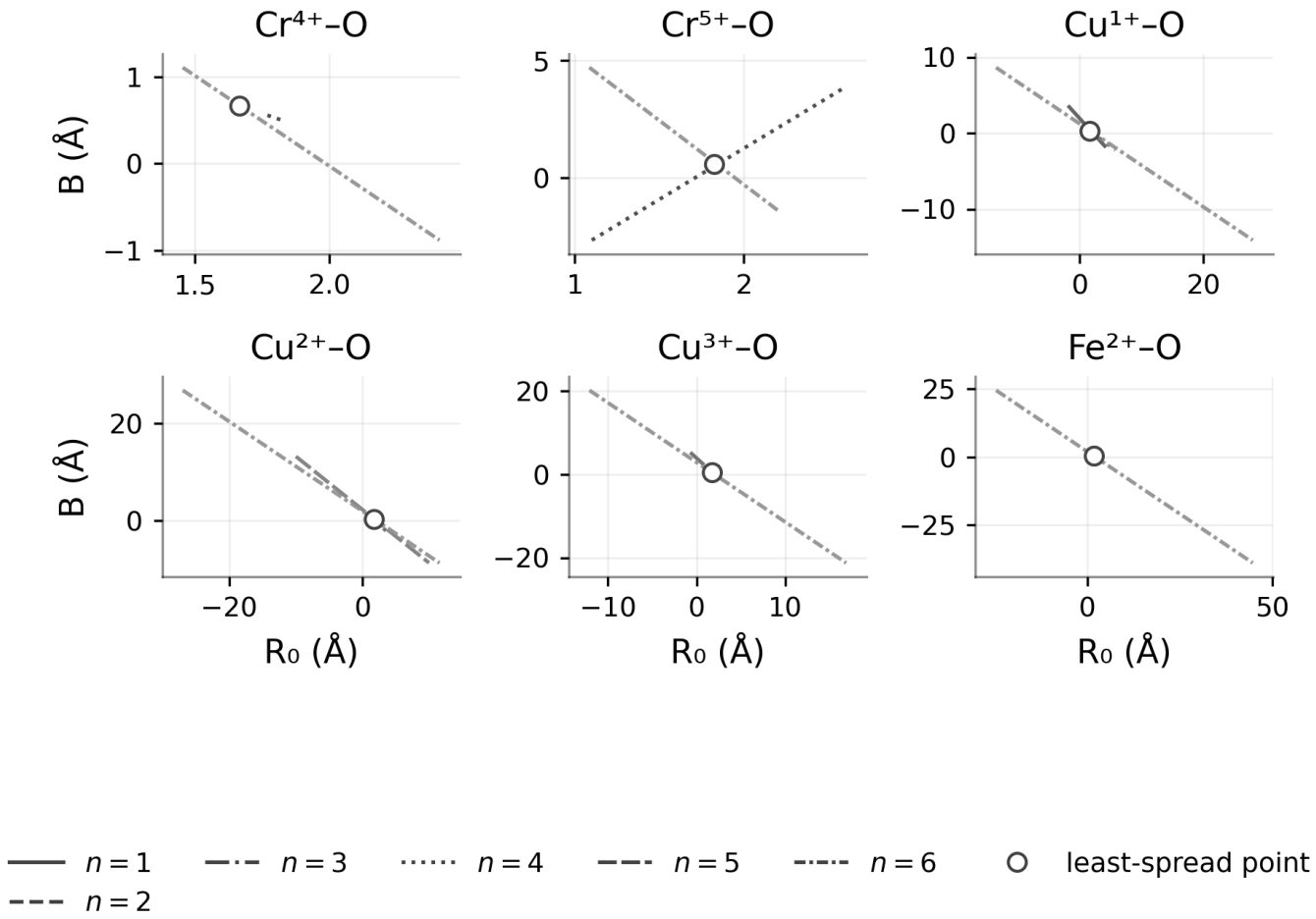

FIG. 4. n-resolved B vs. R_0 for oxidation-state–resolved $3d$ X–O pairs (Cr^{5+}–Fe^{4+}).

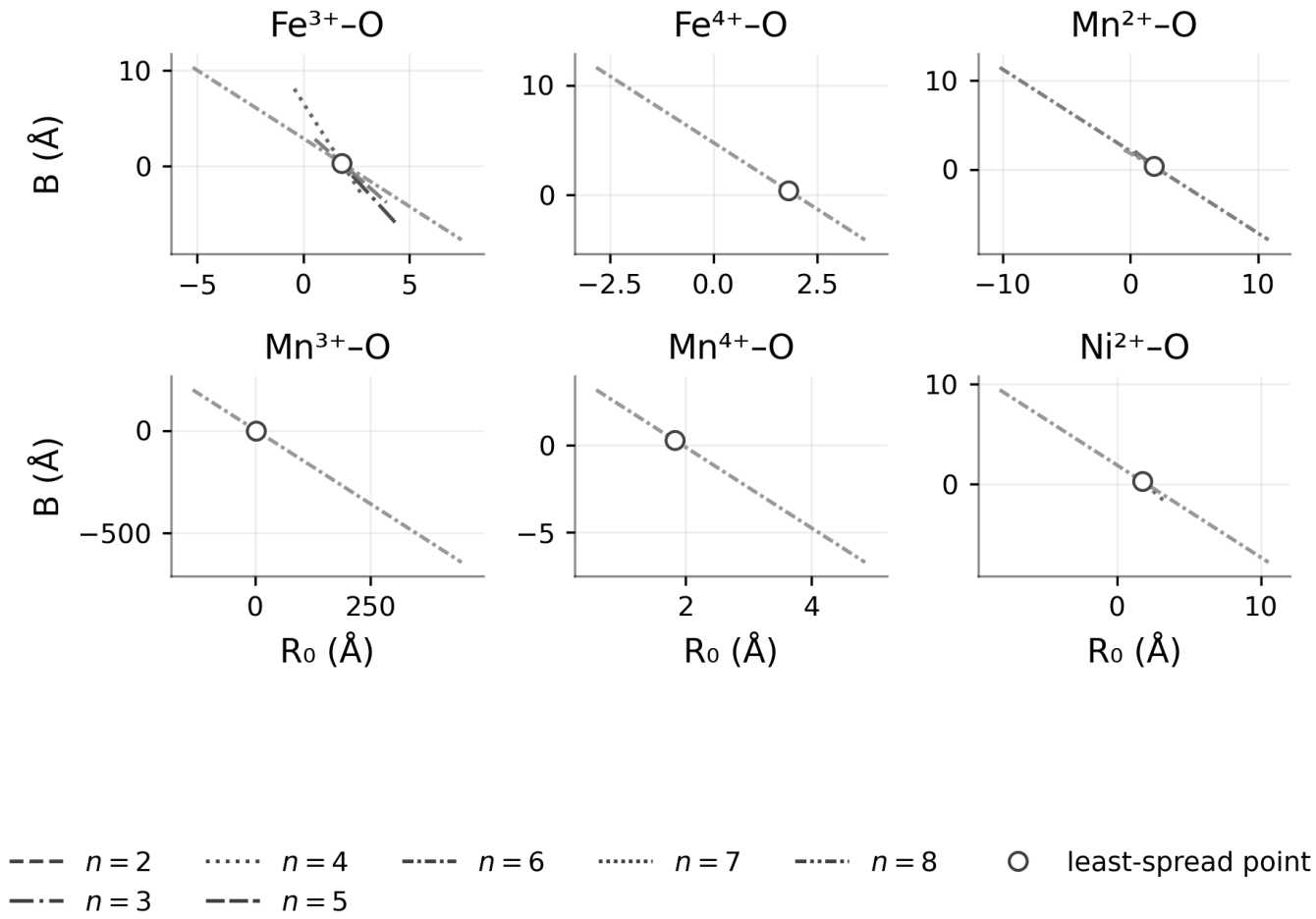

FIG. 5. n-resolved B vs. R_0 for oxidation-state–resolved $3d$ X–O pairs (Co–Cu^{2+}).

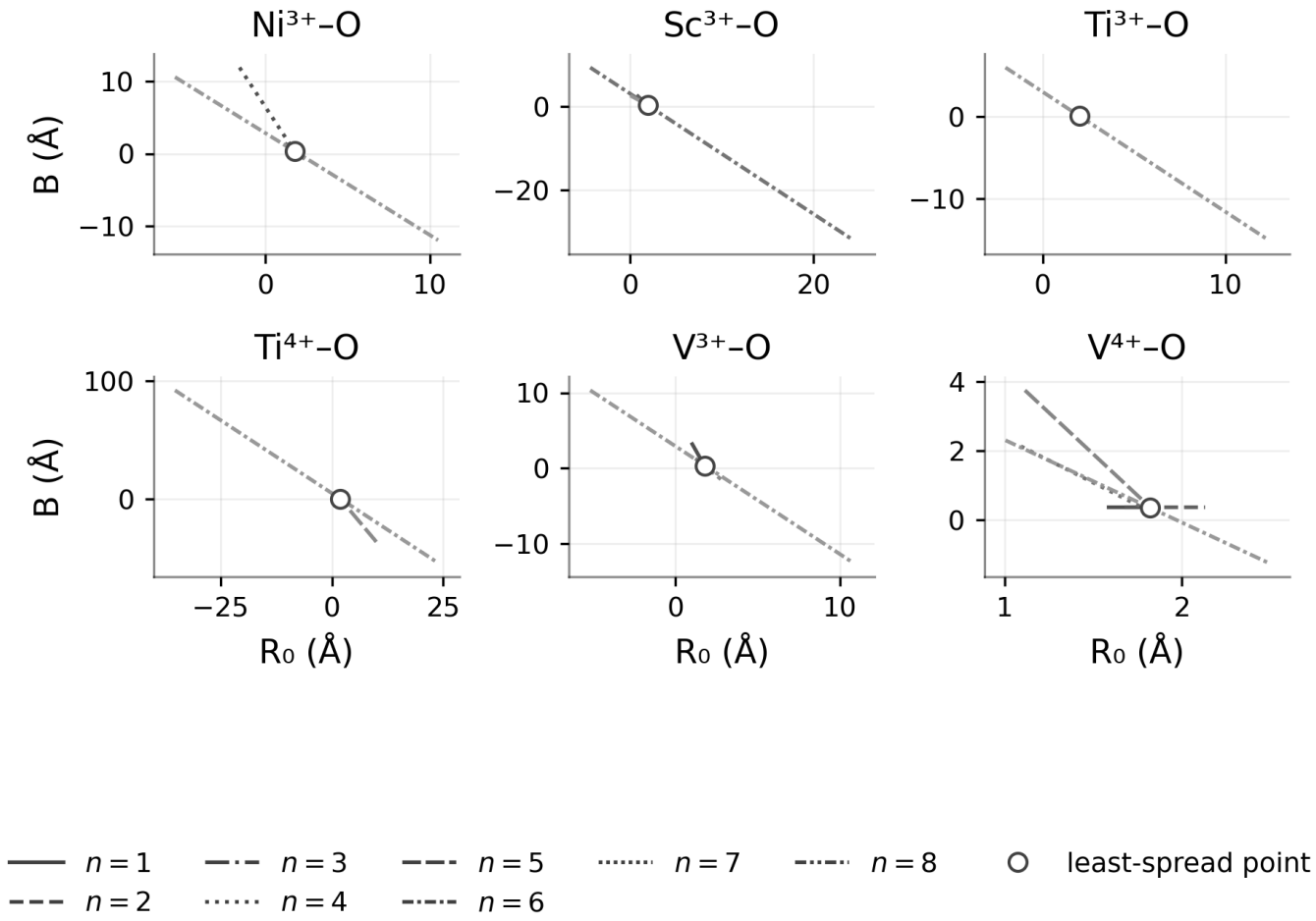

FIG. 6. n-resolved B vs. R_0 for oxidation-state–resolved $3d/4d$ X–O pairs (Cu^{3+}–Nb).

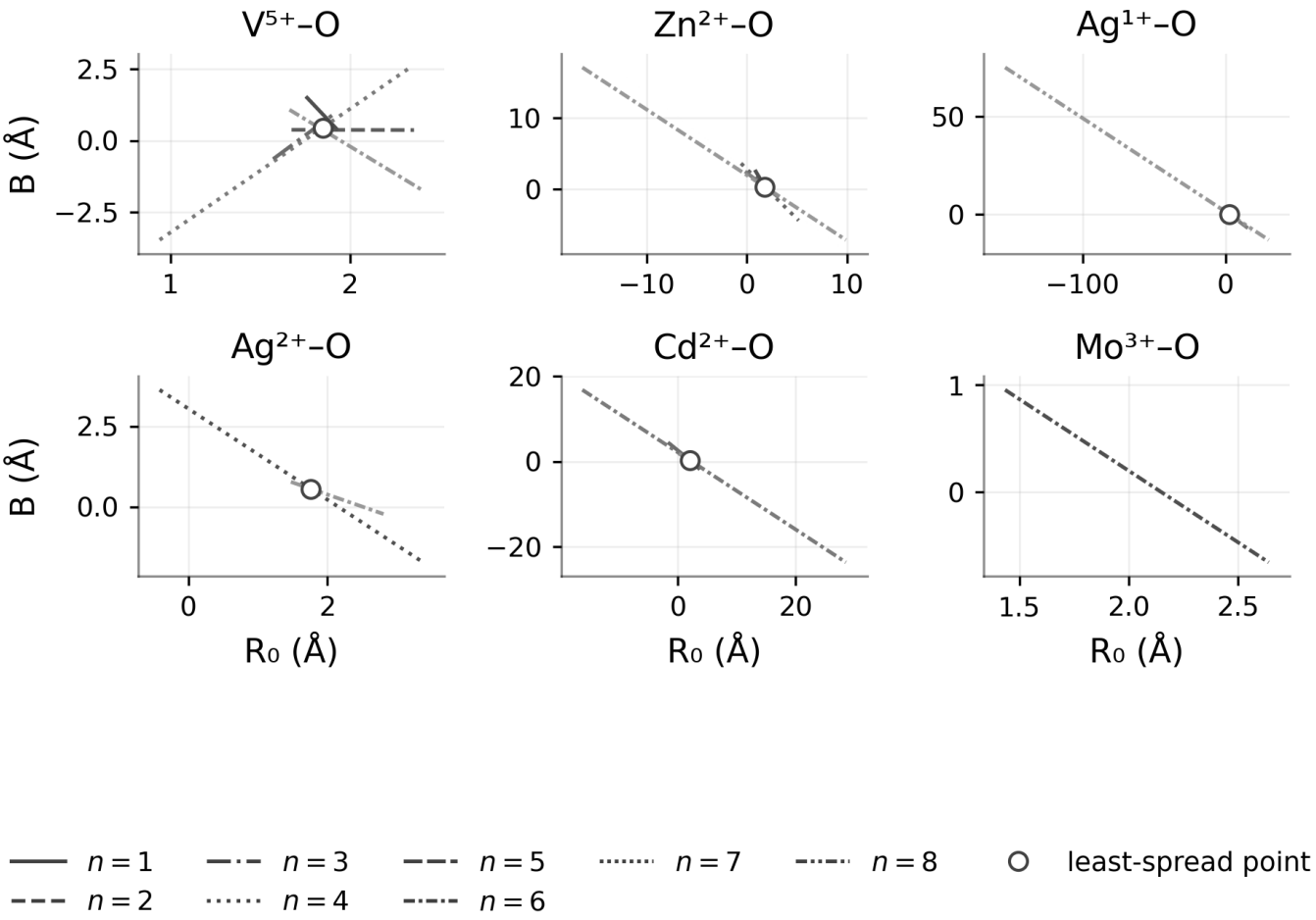

FIG. 7. n-resolved B vs. R_0 for oxidation-state–resolved $4d$ X–O pairs (Mo–Ru).

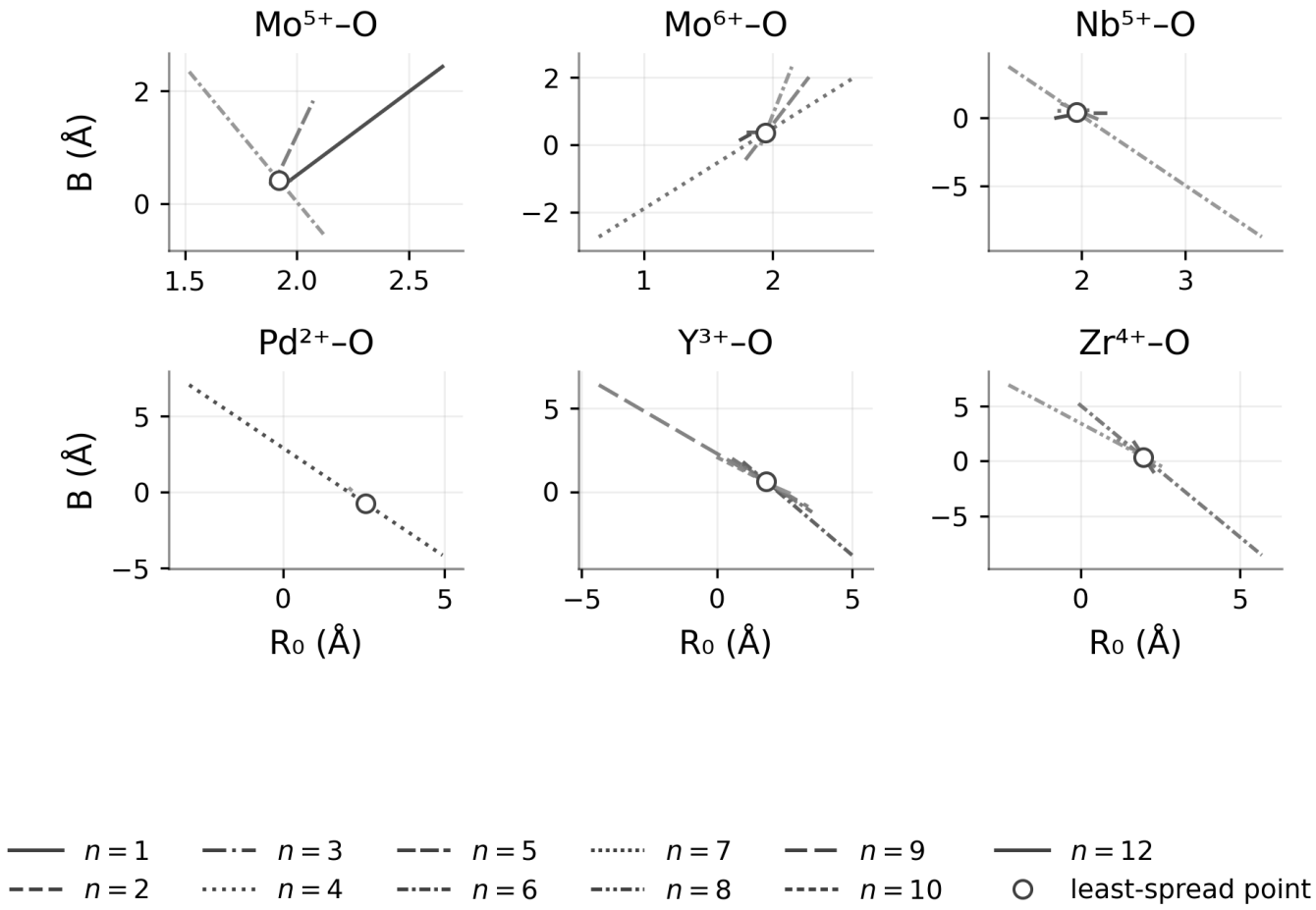

FIG. 8. n-resolved B vs. R_0 for oxidation-state–resolved $4d/5d$ X–O pairs (Rh–Ta^{4+}).

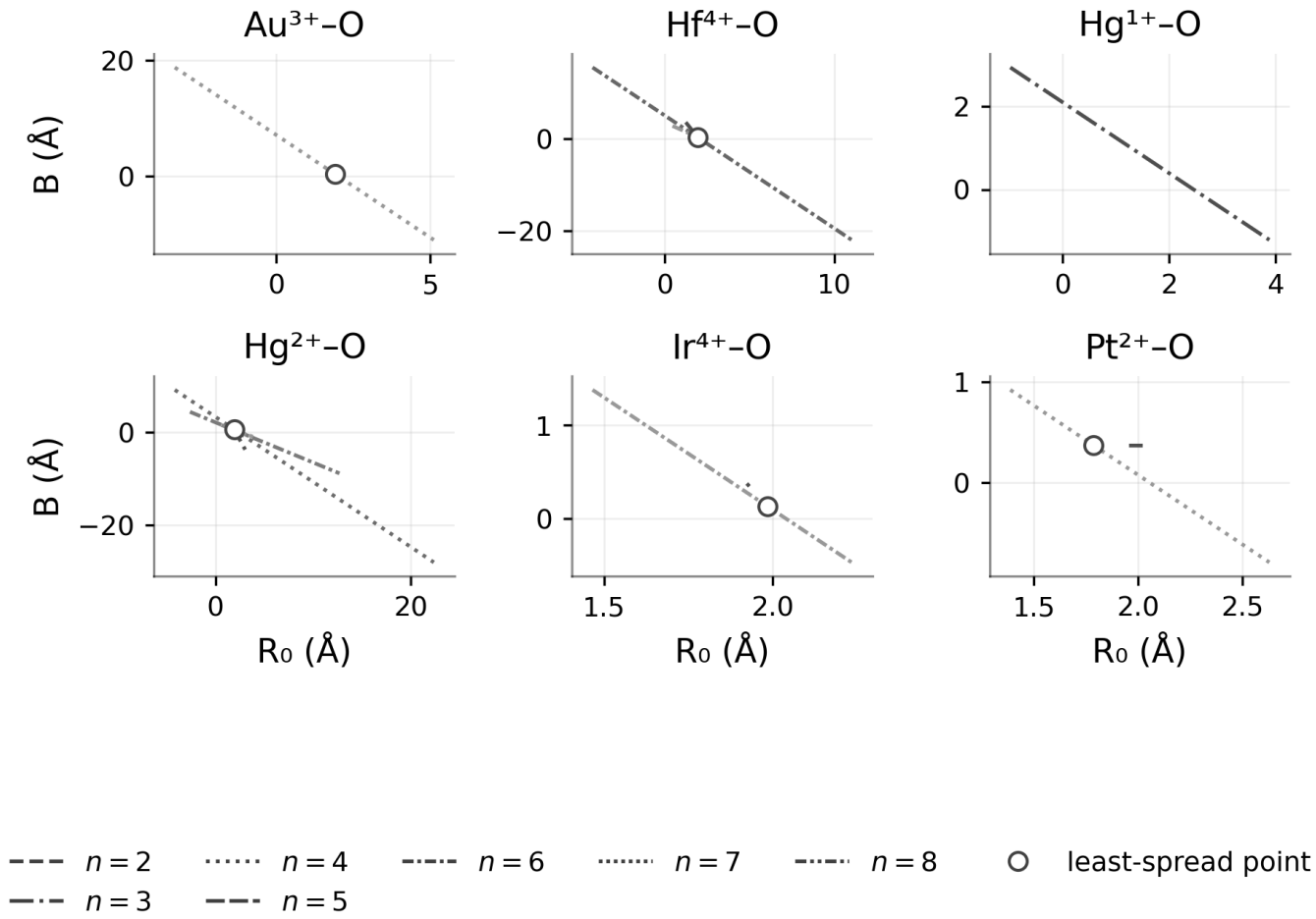

FIG. 9. n-resolved B vs. R_0 for oxidation-state–resolved $5d$ X–O pairs (Ta^{5+}–Os).

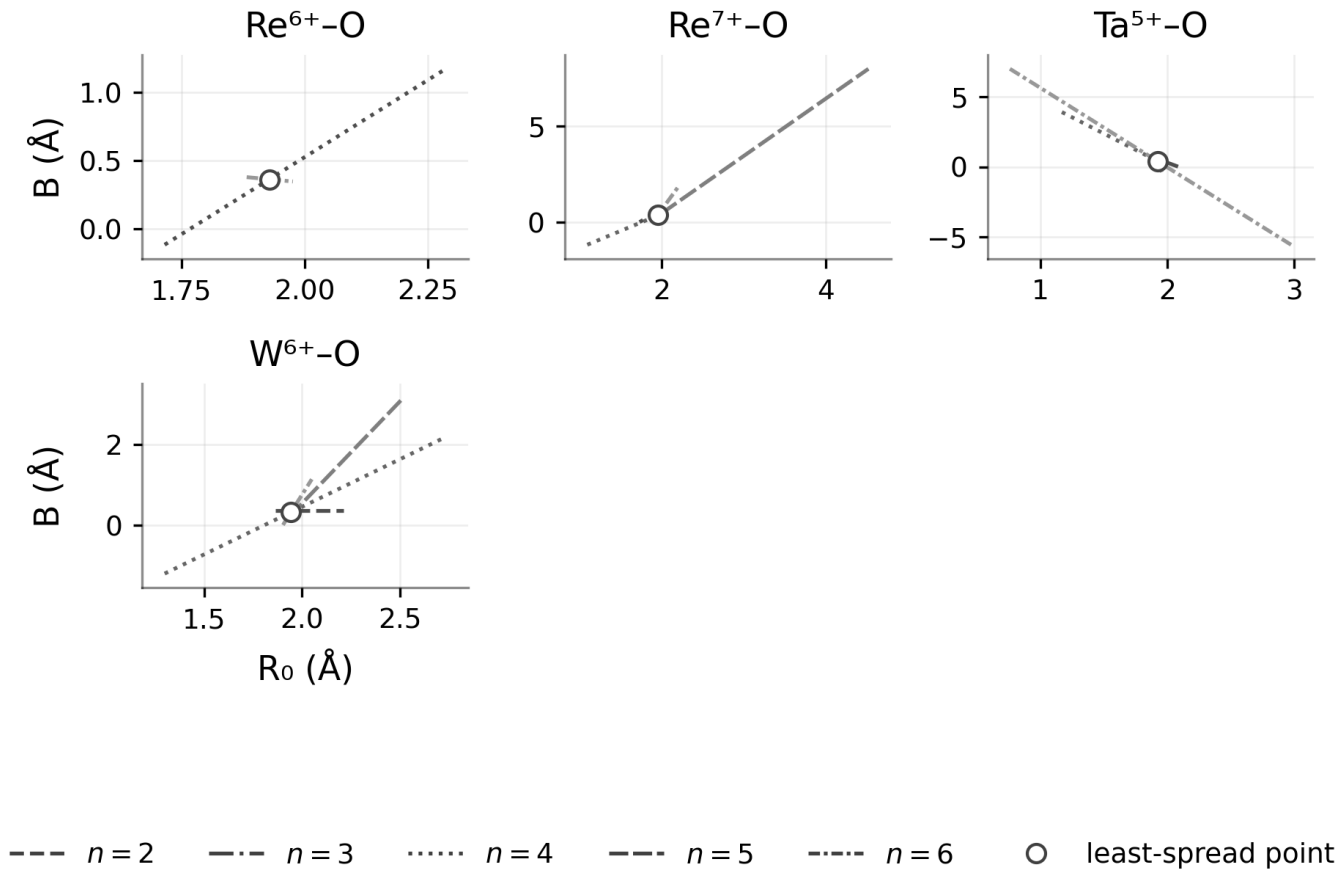

FIG. 10. n-resolved B vs. R_0 for oxidation-state–resolved $5d$ X–O pairs (Ir–Hg).

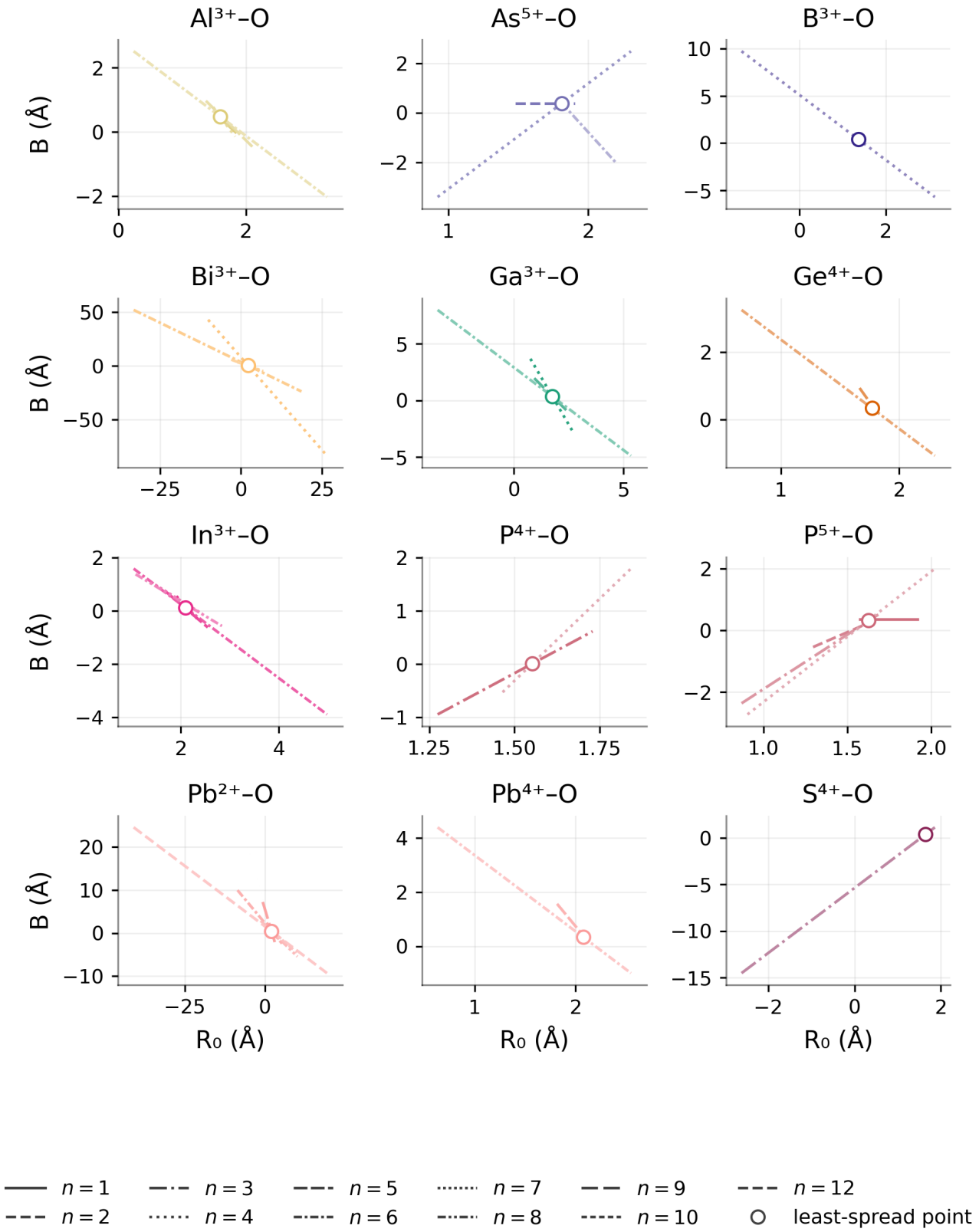

FIG. 11. n-resolved B vs. R_0 for oxidation-state–resolved p-block X–O pairs (Al^{3+}–S^{4+}).

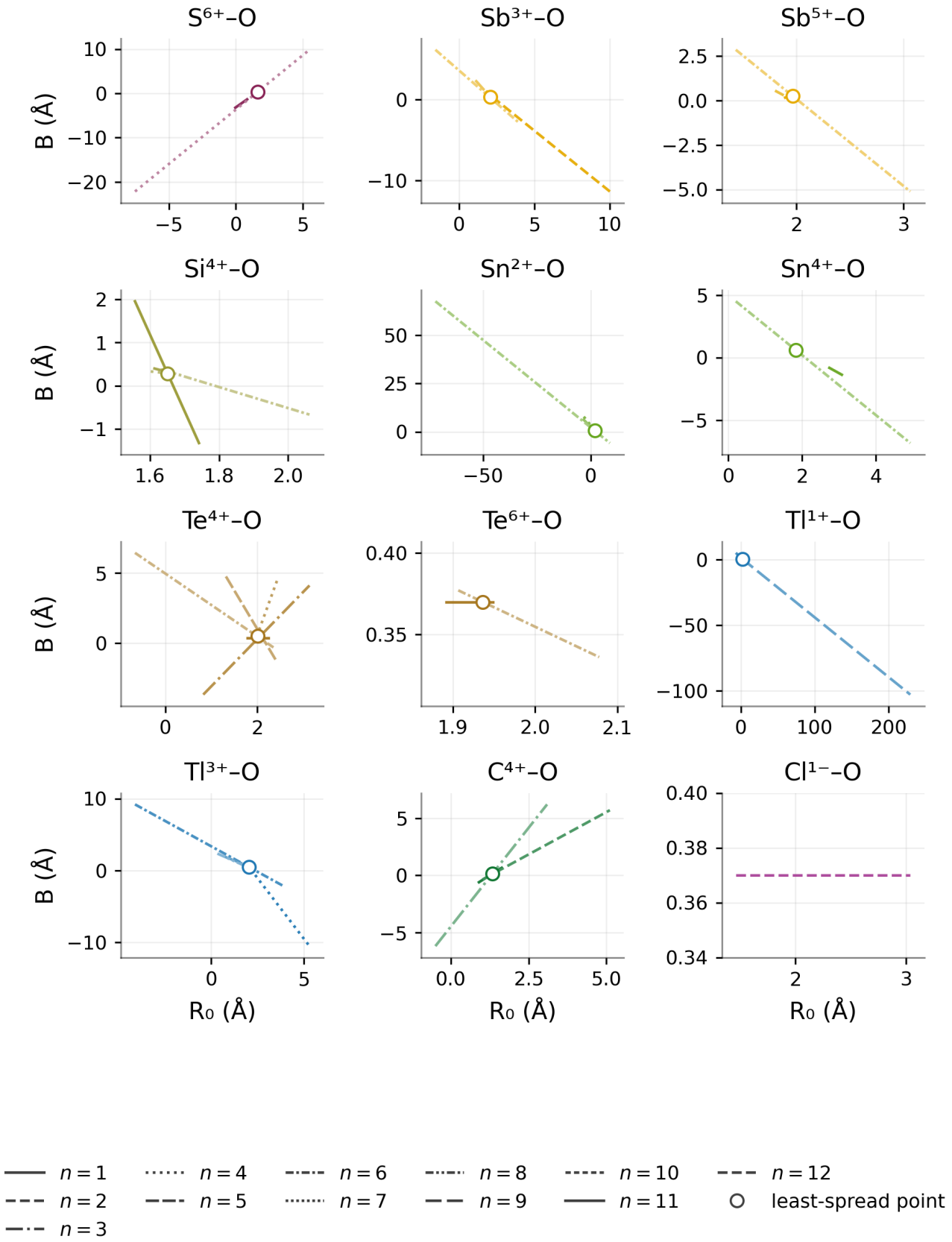

FIG. 12. n-resolved B vs. R_0 for oxidation-state–resolved p-block X–O pairs (S^{6+}–Cl).

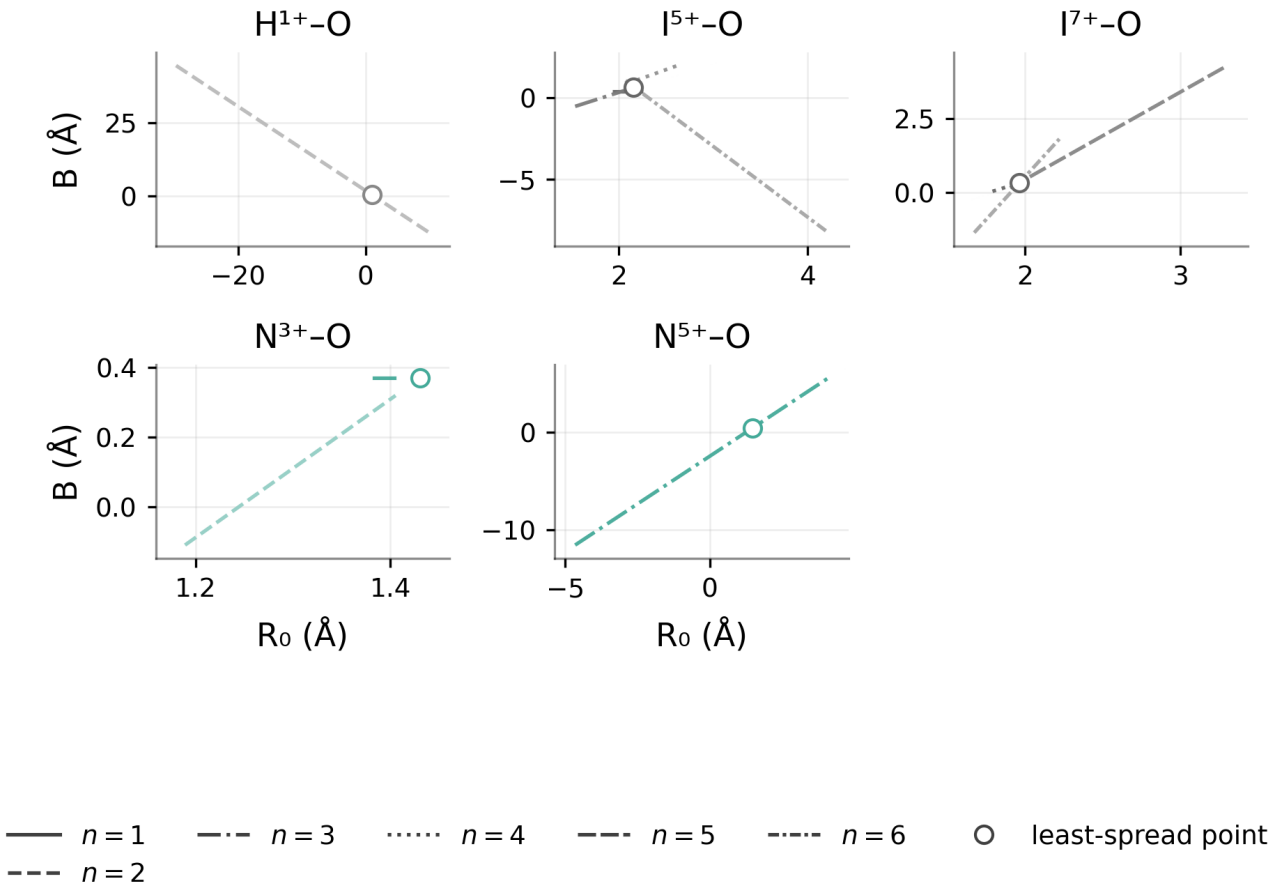

FIG. 13. n-resolved B vs. R_0 for the remaining p-block and hydrogen X–O pairs (H^{1+}, N^{3+}, N^{5+}, I^{5+}, I^{7+}).

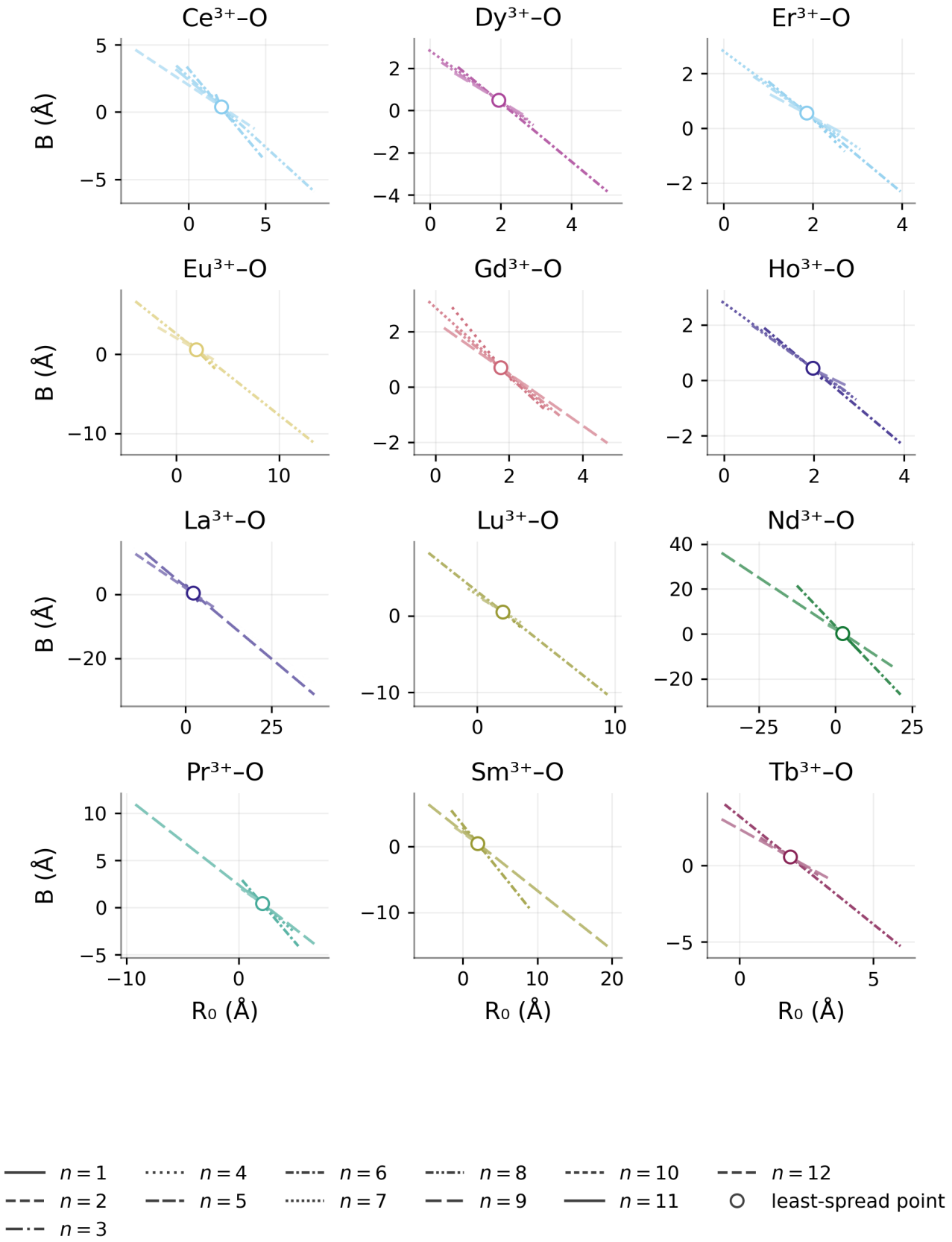

FIG. 14. n-resolved B vs. R_0 for f-block trivalent lanthanide X–O pairs (Ce^{3+}–Tb^{3+}).

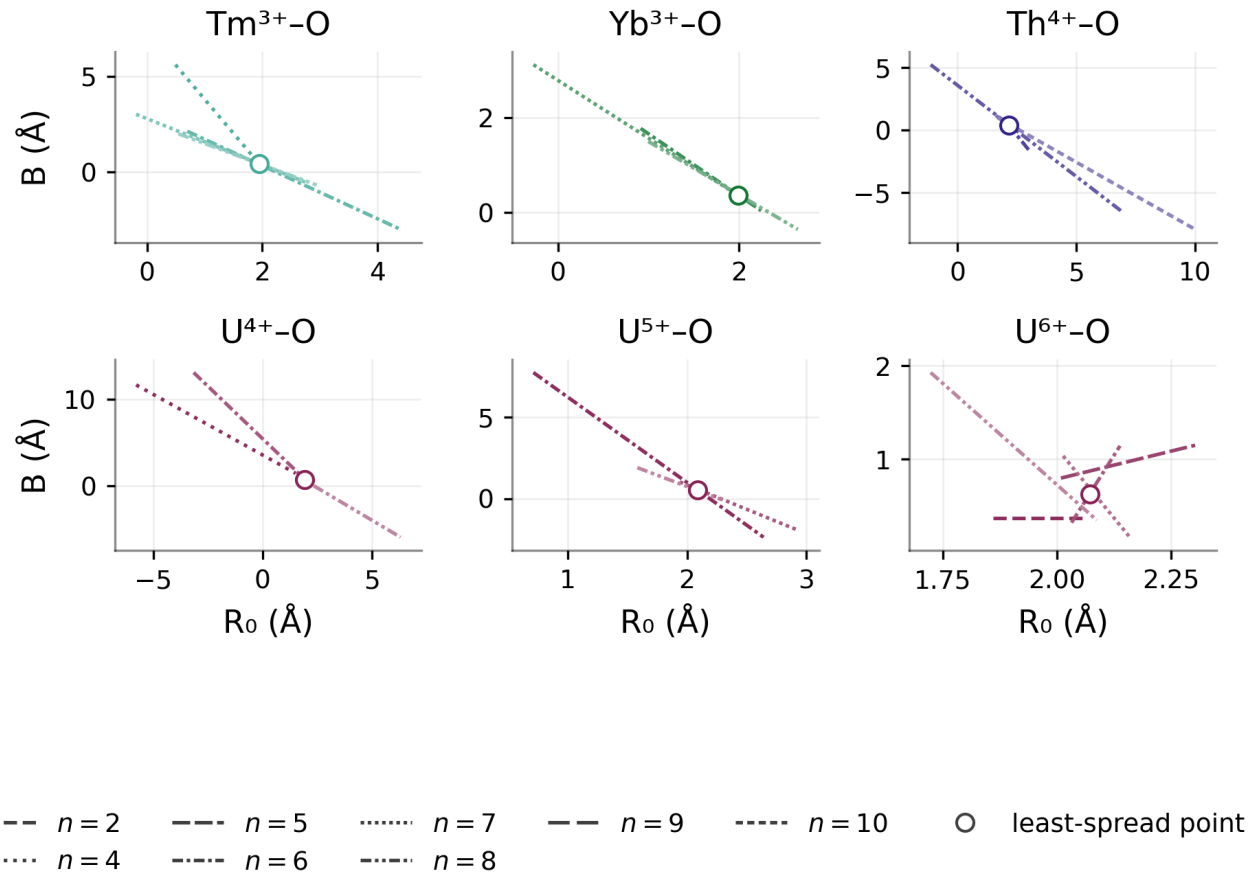

FIG. 15. n-resolved B vs. R_0 for the remaining f-block lanthanide and actinide X–O pairs (Tm^{3+}–Yb^{3+} and Th^{4+}–U^{6+}).

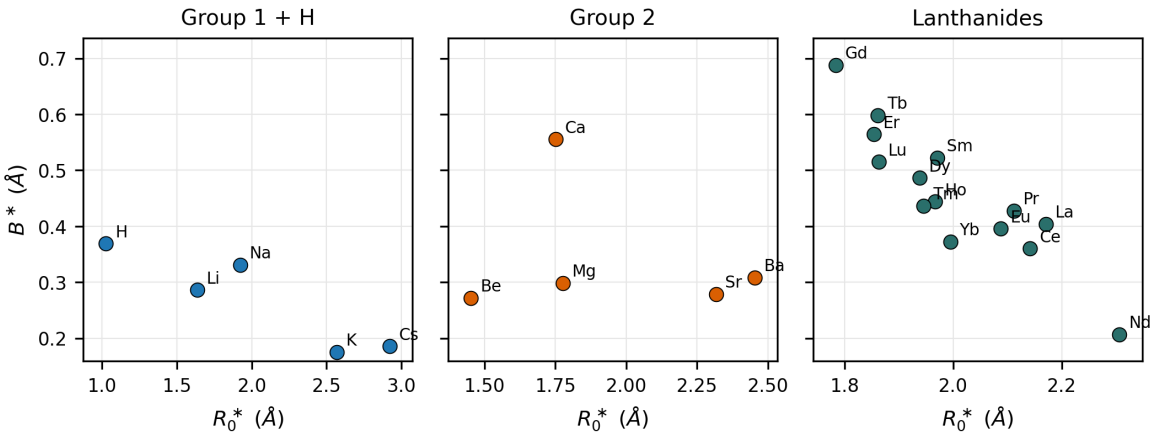

FIG. 16. Characteristic (R_0^*, B^*) for Group 1 including H^+ (left), Group 2 (center), and trivalent lanthanides (right).

CHARACTERISTIC PAIRS AND BOND-LENGTH IDENTITY

Figure 16 shows the characteristic pairs (R_0^*, B^*) for Group 1 (including H^+), Group 2, and the trivalent lanthanides.

For each species and coordination shell n, the fitted points define a line $B = \beta^{(n)} R_0 + \beta_0^{(n)}$. The characteristic pair is the weighted least-spread intersection of those n-resolved lines. If the final line fit for shell n contains N_n structures, we assign weight $w_n = N_n$ and define the weighted means

$$\bar{\beta} = \frac{\sum_n w_n \beta^{(n)}}{\sum_n w_n}, \qquad \bar{\beta}_0 = \frac{\sum_n w_n \beta_0^{(n)}}{\sum_n w_n}.$$

The characteristic abscissa is then

$$R_0^* = -\frac{\sum_n w_n \big(\beta^{(n)} - \bar{\beta}\big)\big(\beta_0^{(n)} - \bar{\beta}_0\big)}{\sum_n w_n \big(\beta^{(n)} - \bar{\beta}\big)^2},$$

which is the point that minimizes the weighted variance of the line-by-line values $B_n(R_0) = \beta^{(n)} R_0 + \beta_0^{(n)}$. The characteristic ordinate and its spread are

$$B^* = \frac{\sum_n w_n B_n(R_0^*)}{\sum_n w_n}, \qquad \sigma_B = \left[\frac{\sum_n w_n \big(B_n(R_0^*) - B^*\big)^2}{\sum_n w_n}\right]^{1/2}.$$

The tabulated n_{CN} counts how many distinct coordination shells contribute to that intersection, while the weights entering the intersection are the fitted structure counts in those shells.

Figure 17 validates the bond-length identity $R_0 \approx \bar{R} + B \ln(z/n)$ across 44,844 screening-branch cation–oxygen structures (records with $B > 0$, $0 < R_0 < 4$ Å, and $z \neq n$; Pearson $r = 0.976$, MAE$= 0.019$ Å).

s-BLOCK SCREENING-CENTROID CONTROLS

For the ten Group 1/Group 2 ground-state binary oxides used in Fig. 2 of the main text, the DFT screening centroid r_{eff} is compared against three linear descriptors: the bond-valence shell centroid m_i, the fitted bond-valence intercept R_0, and the Shannon cation radius. Leave-one-out mean absolute errors are 0.006 Å for m_i, 0.078 Å for R_0, and 0.095 Å for the Shannon-radius control, showing that the r_{eff}–m_i collapse is substantially sharper than a generic monotone size trend.

Each oxide contributes one point (m_i, r_{eff}) to Fig. 2. For a given oxide, the bond-valence shell centroid is computed from the same fitted (R_0, B) and coordination number n used for its bond-valence record, taking the n shortest M–O distances in the ground-state structure and forming

$$p_{ij} = \frac{e^{-R_{ij}/B}}{\sum_{k=1}^{n} e^{-R_{ik}/B}}, \qquad m_i = \sum_{j=1}^{n} p_{ij} R_{ij}.$$

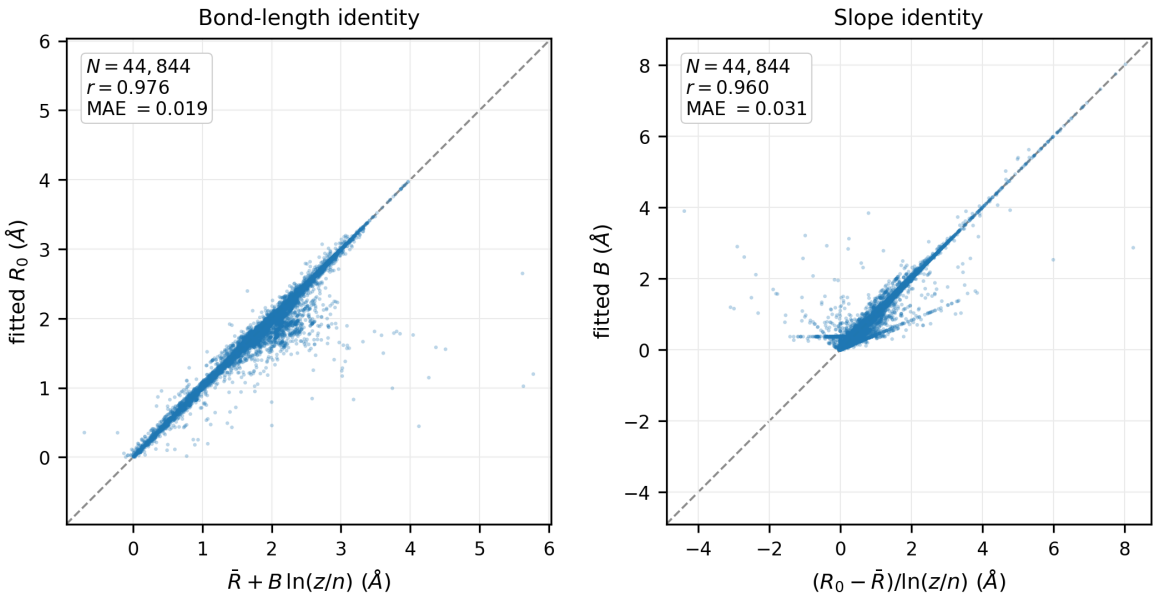

FIG. 17. Left: parity between fitted R_0 and $\bar{R} + B\ln(z/n)$ for 44,844 screening-branch cation–oxygen structures. Right: parity between fitted B and $(R_0 - \bar{R})/\ln(z/n)$ for the same set.

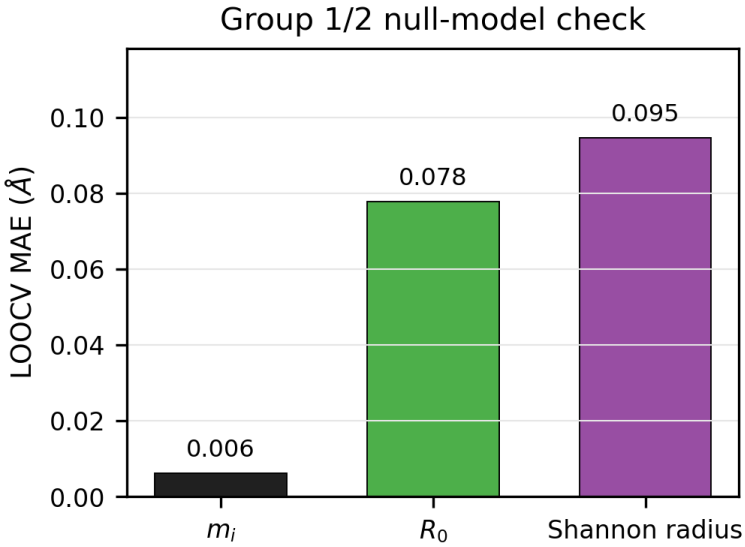

FIG. 18. Leave-one-out null-model check for the ten Group 1/Group 2 binary oxides used in Fig. 2 of the main text. Bars report the mean absolute error of linear regressions against the bond-valence shell centroid m_i, the fitted intercept R_0, and the Shannon cation radius.

The DFT-side centroid is then built bond by bond over all oxygen neighbors with $R_{ij} \leq 3.5$ Å. Each such bond contributes one oxygen-side path centroid $r_{\mathrm{eff}}^{(\mathrm{path})}$, and the oxide-level r_{eff} plotted in Fig. 2 is the simple arithmetic mean of those finite bond-path centroids.

The offset in the main-text regression is quantified directly from that same discrete bond-path construction. Defining the oxygen-side pre-cusp valence centroid shift by

$$\Delta_{\mathrm{O,val}} := \langle R_{ij} - r_{\mathrm{eff}}^{(\mathrm{path})} \rangle,$$

let $\{r_g, \rho_g\}_{g=1}^{200}$ denote the total-density samples on the nearest-image straight M–O segment at 200 uniformly spaced points. The bond critical-point proxy is the interior density minimum on the interval $0.15R_{ij} < r < 0.85R_{ij}$; denote its location and value by r_{BCP} and ρ_{BCP}. The oxygen-side pre-cusp window is the subset of sampled points with $r_{\mathrm{BCP}} \leq r_g \leq 0.93R_{ij}$, finite ρ_g, and positive finite Thomas–Fermi screening length $\lambda_{\mathrm{TF}}(\rho_g)$. The Thomas–Fermi

length enters here only as a positivity filter on the sample window and does not appear in the centroid weights or in any of the regression results; the `thomas_fermi` tag in the Fig. 2 file label reflects only the screening interpretation of $r_{\rm eff}$, not its construction. On that discrete window we use the excess-density weights $w_g = \max[\rho_g - \rho_{\rm BCP}, 0]$; in the rare case that these clipped weights vanish identically, we use $w_g = \rho_g$ instead. Thus

$$r_{\rm eff}^{(\rm path)} = \frac{\sum_g w_g r_g}{\sum_g w_g},$$

and the oxide-level centroid used in Fig. 2 is the arithmetic mean of $r_{\rm eff}^{(\rm path)}$ over those first-shell M–O bonds. No spline fitting, path relaxation, or topological bond-path search is introduced beyond this fixed straight-segment sampling rule. With that definition, the ten-oxide s-block shift fit is $\Delta_{\rm O,val} = 0.0613\, m_i + 0.2394$ Å ($R^2 = 0.963$), with mean 0.3910 ± 0.0269 Å. Equivalently, $r_{\rm eff} = 0.9387\, m_i - 0.2394$ Å, whose rounded form is the main-text relation. The intercept is therefore the zero-radius limit of a directly measured oxygen-side screening shift, not an unconstrained empirical parameter.

Species-level screening length from the characteristic pair. The shell-level Yukawa inversion in the main text is written in terms of the representative radius R^*, chosen operationally as the Boltzmann shell centroid m_i. To pass from that shell-level relation to a species-level constant, we evaluate the inversion at the characteristic fixed point (R_0^*, B^*) on the characteristic reference shell defined by $m_i = R_0^*$. Using the exact identity

$$R_0 = m_i + B(\ln q_i - H_i),$$

this reference-shell condition is equivalent at the fixed point to $H_i = \ln q_i$, so the representative radius entering the Yukawa inversion is $R^* = R_0^*$. Substituting $B = B^*$ and $R^* = R_0^*$ into the positive screening branch therefore gives

$$\lambda^* = \frac{-R_0^* + \sqrt{R_0^{*2} + 4B^* R_0^*}}{2},$$

which is the species-level screening length reported in Table I for entries with $B^* > 0$.

SCREENING LENGTHS FOR THE ANALYZED SPECIES

TABLE I: Characteristic pairs (R_0^*, B^*) for all 103 cation–oxygen species and physical screening lengths λ^* for the 101 screening-branch species, ordered by atomic number. All values are in Å. For entries with positive characteristic ordinate ($B^* > 0$), λ^* is computed from the screening-branch inversion in the main text; $n_{\rm CN}$ is the number of distinct coordination shells used to determine (R_0^*, B^*). Parenthetical uncertainties on B^* are the weighted standard deviation σ_B of the n-resolved lines at R_0^*; uncertainties on λ^* are propagated for entries with $B^* > 0$ via $\sigma_\lambda = |\partial\lambda^*/\partial B^*|\,\sigma_B$. Entries with $n_{\rm CN} = 2$ have $\sigma_B = 0$ by construction and should be regarded as provisional. Two species (Mo^{3+} and Sb^{3+}) yield $B^* < 0$; these are retained as signed Yukawa-branch diagnostics that the first-order expansion does not adequately describe their bonding and are not assigned physical λ^* values.

Species	Block	R_0^*	B^*	λ^*	$n_{\rm CN}$
H^{1+}	s	1.026	0.369(9)	0.288(6)	9
Li^{1+}	s	1.636	0.286(22)	0.248(17)	10
Be^{2+}	s	1.452	0.272(0)	0.234(0)	2
B^{3+}	p	1.376	0.372(9)	0.304(6)	7
C^{4+}	p	1.337	0.135(4)	0.123(3)	3
N^{3+}	p	1.431	0.370(0)	0.305(0)	2
N^{5+}	p	1.492	0.437(0)	0.354(0)	2
Na^{1+}	s	1.923	0.331(15)	0.288(11)	10
Mg^{2+}	s	1.776	0.298(16)	0.260(12)	5
Al^{3+}	p	1.648	0.407(9)	0.338(6)	6
Si^{4+}	p	1.652	0.269(9)	0.235(7)	5
P^{4+}	p	1.552	0.004(0)	0.004(0)	2

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TABLE I – continued from previous page

Species	Block	R_0^*	B^*	λ^*	n_{CN}
P^{5+}	p	1.625	0.324(7)	0.277(6)	4
S^{4+}	p	1.644	0.370(0)	0.311(0)	3
S^{6+}	p	1.610	0.307(0)	0.264(0)	2
Cl^{1-}	p	1.804	0.370(0)	0.315(0)	2
K^{1+}	s	2.568	0.175(24)	0.165(21)	12
Ca^{2+}	s	1.751	0.555(35)	0.443(24)	8
Sc^{3+}	d	1.961	0.222(19)	0.201(16)	6
Ti^{3+}	d	2.009	0.080(0)	0.077(0)	2
Ti^{4+}	d	1.881	0.248(43)	0.222(35)	6
V^{3+}	d	1.782	0.375(22)	0.318(16)	6
V^{4+}	d	1.821	0.353(15)	0.303(11)	5
V^{5+}	d	1.849	0.448(75)	0.373(53)	6
Cr^{2+}	d	1.744	0.435(14)	0.360(10)	3
Cr^{3+}	d	1.761	0.370(0)	0.314(0)	3
Cr^{4+}	d	1.665	0.670(0)	0.512(0)	2
Cr^{5+}	d	1.821	0.544(93)	0.439(63)	3
Mn^{2+}	d	1.830	0.354(6)	0.304(5)	7
Mn^{3+}	d	1.810	0.357(18)	0.305(13)	7
Mn^{4+}	d	1.830	0.296(0)	0.260(0)	2
Fe^{2+}	d	1.760	0.385(22)	0.325(16)	6
Fe^{3+}	d	1.799	0.352(9)	0.302(6)	5
Fe^{4+}	d	1.816	0.394(0)	0.333(0)	2
Co^{1+}	d	1.584	0.370(0)	0.310(0)	2
Co^{2+}	d	1.712	0.384(6)	0.323(5)	4
Co^{3+}	d	1.776	0.354(5)	0.303(3)	4
Co^{4+}	d	1.774	0.330(38)	0.284(29)	3
Ni^{2+}	d	1.766	0.304(8)	0.264(6)	5
Ni^{3+}	d	1.773	0.364(12)	0.310(9)	3
Cu^{1+}	d	1.585	0.319(53)	0.273(40)	4
Cu^{2+}	d	1.715	0.358(36)	0.304(27)	7
Cu^{3+}	d	1.708	0.481(42)	0.391(29)	3
Zn^{2+}	d	1.760	0.328(7)	0.283(5)	4
Ga^{3+}	p	1.756	0.357(5)	0.304(4)	4
Ge^{4+}	p	1.771	0.354(15)	0.302(11)	4
As^{5+}	p	1.812	0.382(1)	0.324(1)	3
Rb^{1+}	s	2.809	0.146(16)	0.140(15)	10
Sr^{2+}	s	2.317	0.279(17)	0.252(14)	9
Y^{3+}	d	1.806	0.634(18)	0.497(12)	9
Zr^{4+}	d	1.979	0.330(27)	0.288(21)	6
Nb^{5+}	d	1.945	0.426(30)	0.359(22)	7
Mo^{3+}	d	2.324	−0.237(0)	—	2
Mo^{5+}	d	1.919	0.415(46)	0.351(34)	4
Mo^{6+}	d	1.948	0.368(19)	0.317(14)	5
Pd^{2+}	d	1.402	0.918(0)	0.632(0)	2
Ag^{1+}	d	2.477	0.011(61)	0.011(60)	8
Ag^{2+}	d	1.778	0.548(0)	0.439(0)	2
Cd^{2+}	d	2.083	0.247(18)	0.223(15)	6
In^{3+}	p	2.065	0.162(34)	0.151(30)	5
Sn^{2+}	p	1.903	0.607(45)	0.484(30)	4
Sn^{4+}	p	1.845	0.567(9)	0.455(6)	3
Sb^{3+}	p	2.434	−0.341(87)	—	5
Sb^{5+}	p	1.990	0.130(39)	0.122(35)	4
Te^{4+}	p	2.023	0.562(151)	0.458(104)	7
Te^{6+}	p	1.937	0.370(0)	0.318(0)	2
I^{5+}	p	2.077	0.476(34)	0.399(24)	6
I^{7+}	p	1.964	0.324(4)	0.283(3)	3
Cs^{1+}	s	2.921	0.185(21)	0.175(19)	11
Ba^{2+}	s	2.453	0.308(11)	0.277(9)	10
La^{3+}	f	2.171	0.404(41)	0.348(31)	12

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TABLE I – continued from previous page

Species	Block	R_0^*	B^*	λ^*	n_{CN}
Ce^{3+}	*f*	2.142	0.360(50)	0.314(39)	9
Pr^{3+}	*f*	2.112	0.427(30)	0.364(22)	7
Nd^{3+}	*f*	2.307	0.207(50)	0.191(43)	8
Sm^{3+}	*f*	1.971	0.522(13)	0.429(9)	8
Eu^{3+}	*f*	2.088	0.396(28)	0.340(21)	6
Gd^{3+}	*f*	1.784	0.688(28)	0.530(17)	6
Tb^{3+}	*f*	1.862	0.598(18)	0.476(12)	5
Dy^{3+}	*f*	1.938	0.486(18)	0.403(13)	5
Ho^{3+}	*f*	1.967	0.444(16)	0.373(11)	5
Er^{3+}	*f*	1.854	0.564(10)	0.453(7)	5
Tm^{3+}	*f*	1.946	0.436(18)	0.367(13)	4
Yb^{3+}	*f*	1.995	0.372(7)	0.321(5)	4
Lu^{3+}	*f*	1.863	0.515(13)	0.420(9)	5
Hf^{4+}	*d*	1.929	0.362(16)	0.311(12)	4
Ta^{5+}	*d*	1.924	0.400(14)	0.340(10)	5
W^{6+}	*d*	1.936	0.305(20)	0.268(15)	4
Re^{6+}	*d*	1.928	0.363(0)	0.312(0)	2
Re^{7+}	*d*	1.948	0.377(72)	0.323(54)	4
Ir^{4+}	*d*	1.984	0.133(0)	0.125(0)	2
Pt^{2+}	*d*	1.785	0.370(0)	0.315(0)	2
Au^{3+}	*d*	1.924	0.363(13)	0.312(10)	3
Hg^{1+}	*d*	2.230	0.233(28)	0.213(24)	4
Hg^{2+}	*d*	1.929	0.605(122)	0.484(81)	6
Tl^{1+}	*p*	2.459	0.272(19)	0.247(16)	10
Tl^{3+}	*p*	2.054	0.469(42)	0.394(30)	4
Pb^{2+}	*p*	2.240	0.348(31)	0.306(24)	10
Pb^{4+}	*p*	2.080	0.353(23)	0.308(17)	3
Bi^{3+}	*p*	2.104	0.430(31)	0.366(23)	8
Th^{4+}	*f*	2.128	0.459(26)	0.388(19)	5
U^{4+}	*f*	1.938	0.787(91)	0.600(56)	3
U^{5+}	*f*	2.089	0.521(23)	0.431(16)	3
U^{6+}	*f*	2.080	0.603(92)	0.489(63)	7

[1] C. Li, N. Katyal, M. Kunz, P. Zarzycki, L. Leinbach, B. Brugman, A. Navrotsky, and M. L. Whittaker, Lithium pyroxene formation threshold predicted by data-driven global bond valence, American Mineralogist (2026), accepted.